CBPMS / SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Project scheduling - Gantt chart

Proposed academic baseline: 07 September - 29 November 2026 (12 weeks). Bars are planned effort, not actual progress.

ID / WORK PACKAGE		AFTER	W01 07 Sep	W02 14 Sep	W03 21 Sep	W04 28 Sep	W05 05 Oct	W06 12 Oct	W07 19 Oct	W08 26 Oct	W09 02 Nov	W10 09 Nov	W11 16 Nov	W12 23 Nov
01	Scope, requirements & feasibility	-												
02	Privacy, domain & API design	01												
03	Persistent data & private storage	02												
04	Google identity & access control	02												
05	Customer website & document flow	03,04												
06	Shop queue & tenant isolation	03,04												
07	Verified payment integration	03,04												
08	Desktop claim / spool / cleanup	05-07												
09	Integration & failure recovery	08												
10	Security & acceptance testing	09												
11	Deployment & handover	10												
12	Documentation & practical outputs	parallel												
13	Final demonstration & review	10,11												

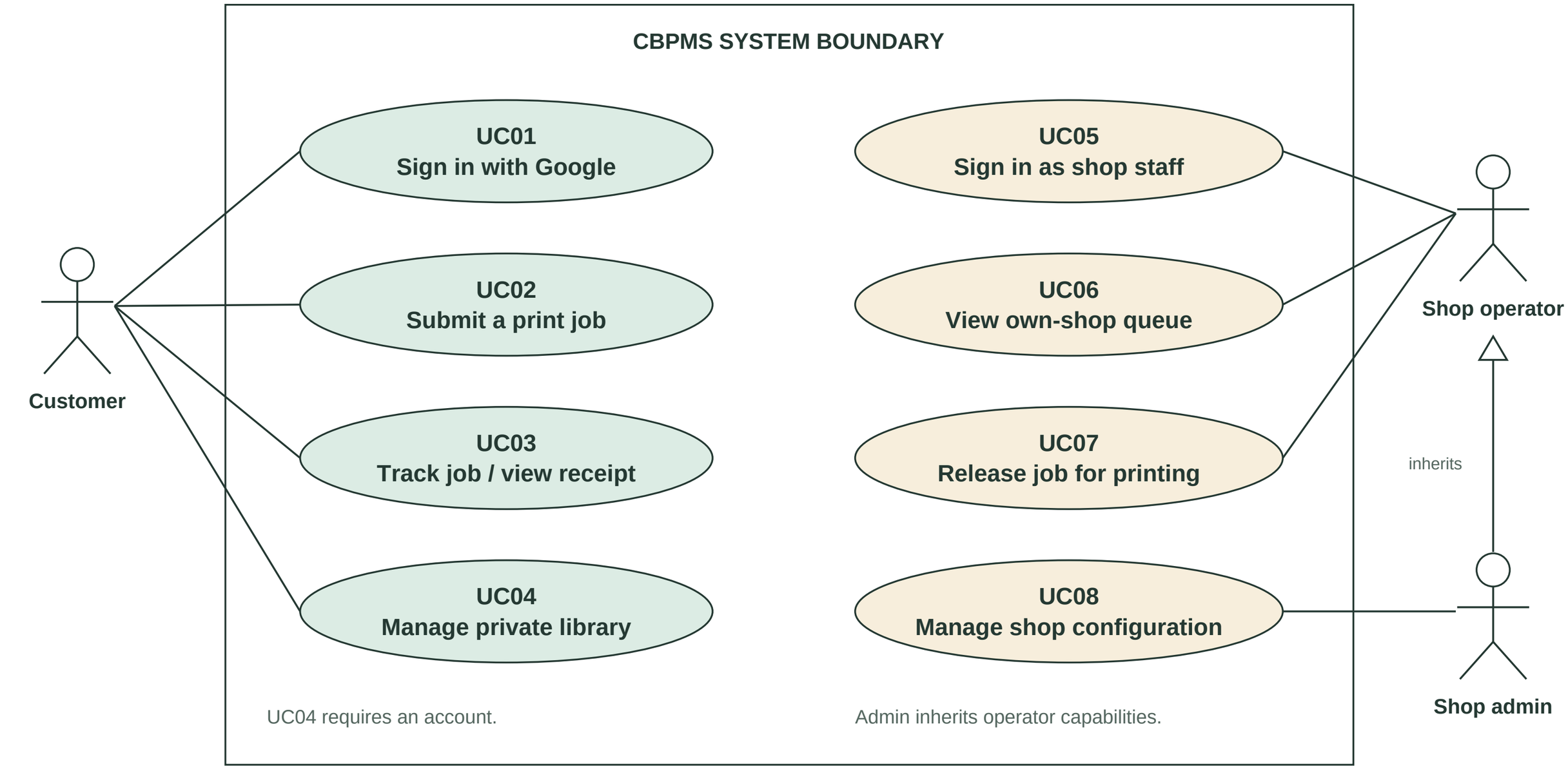
Milestone gates

W04 Identity + data foundation | W07 Web + queue + payment contracts | W11 Acceptance sign-off

Dependencies are finish-to-start gates unless marked parallel. The diamond is the final review milestone at the end of W12, after handover. Dates are illustrative; revise them to match your deadline.

UML use-case diagram

Human actors and system responsibilities. Customer sign-in is optional for anonymous printing.



Associations are solid, undirected lines. The hollow triangle is UML actor generalization; it points to the more general shop-operator role.

Google/Supabase and the planned payment gateway are supporting external services, detailed in the data-flow and sequence diagrams.

Use-case scenarios

Four representative scenarios. IDs match the use-case diagram; exceptions and privacy rules are part of the expected behaviour.

UC01 / Sign in with Google

ACTOR / TRIGGER
Customer selects Continue with Google.

PRECONDITION
Google provider and callback URL are configured.

MAIN FLOW
1. Redirect for Google consent. 2. Exchange the authorization code. 3. Verify identity and establish a secure session. 4. Open the customer workspace.

ALTERNATIVE
Cancelled or invalid authorization returns a safe error; no authenticated session is created.

POSTCONDITION
A returning identity is reused; a new identity gets a customer profile. No staff permissions are granted.

UC02 / Submit a print job

ACTOR / TRIGGER
Customer chooses a shop and starts an upload.

PRECONDITION
Shop is active; account mode requires sign-in.

MAIN FLOW
1. Choose privacy mode. 2. Upload PDF/image. 3. Server validates and prices options. 4. Stage job privately. 5. Verify payment on server. 6. Queue job and issue release PIN.

ALTERNATIVE
Invalid file is rejected. Failed or unavailable payment does not queue the job or issue a release PIN.

POSTCONDITION
Job belongs to one shop. A capability token protects anonymous status. Live payment remains planned.

UC07 / Release job for printing

ACTOR / TRIGGER
Shop operator scans or enters a release PIN.

PRECONDITION
Staff belongs to the job shop; paid job is unexpired.

MAIN FLOW
1. Atomically claim job. 2. Issue a short-lived download grant. 3. Agent downloads and verifies hash. 4. Spool to printer. 5. Report outcome and clear local temporary file.

ALTERNATIVE
Wrong shop, invalid PIN or duplicate claim is denied. Spool failure is reported; retain cloud file for retry until expiry.

POSTCONDITION
Spool success marks completion. Anonymous cloud file is deleted; a reusable library file remains private.

UC04 / Manage private library

ACTOR / TRIGGER
Signed-in customer opens the document library.

PRECONDITION
A valid customer session identifies the owner.

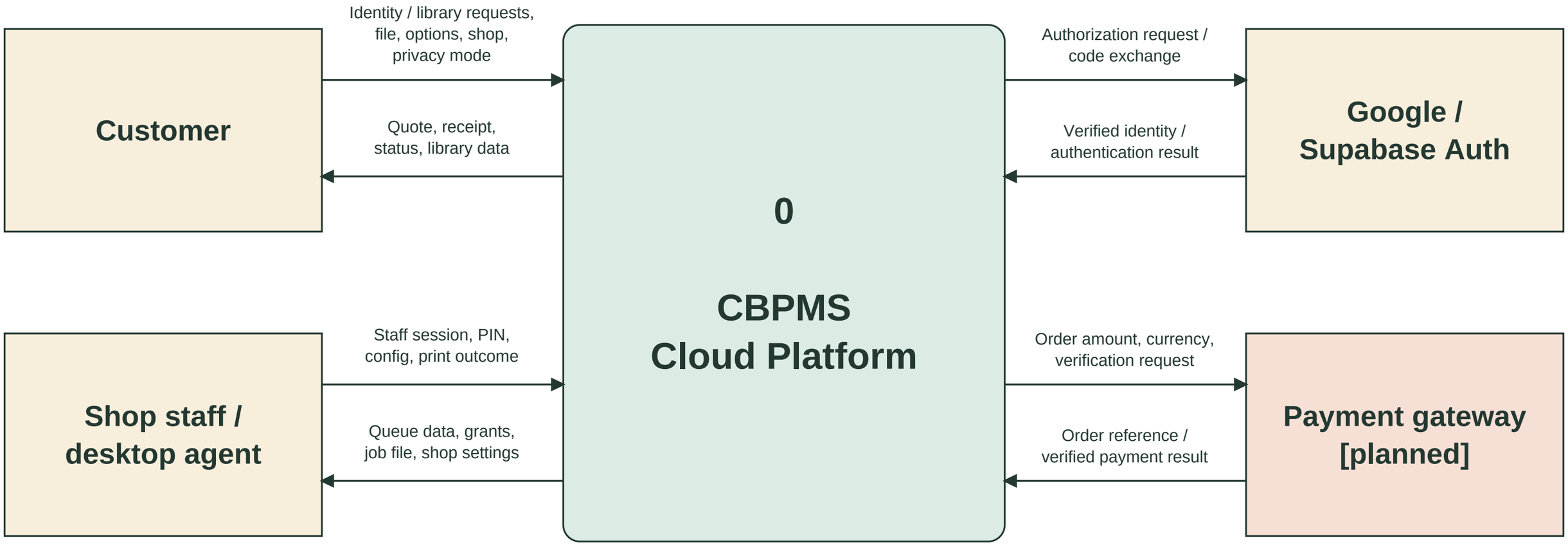
MAIN FLOW
1. List only the owner's account-linked documents. 2. Choose a document to reuse or delete. 3. Check ownership and active jobs. 4. Apply the permitted operation.

ALTERNATIVE
Other owners' files are inaccessible. Deletion during an active print job is blocked; retry after the job finishes.

POSTCONDITION
Reusable documents remain private. Deleted documents become inaccessible. Anonymous uploads never appear here.

Data-flow diagram - Level 0 (context)

Boundary: CBPMS cloud platform (customer/shop web + API + private persistence). Desktop agent is outside this boundary.



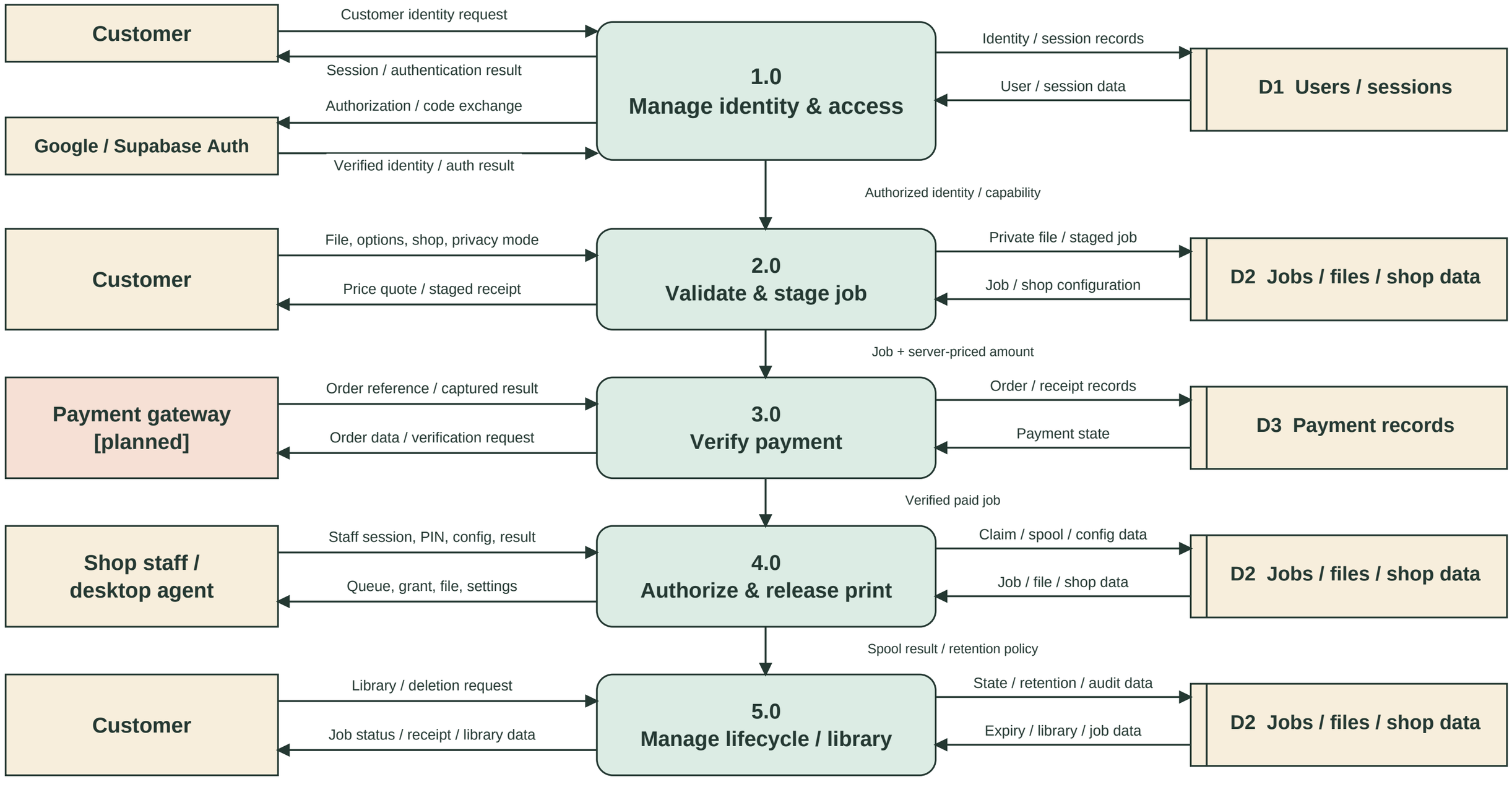
External entities are rectangles; the rounded process represents the entire platform. Arrows carry named data, not execution order.

Internal databases and private object storage are hidden at context level. They appear as data stores in Level 1.

Design scope: live payment integration is proposed; Google authentication and anonymous job mode are part of the current design.

Data-flow diagram - Level 1

Decomposition of process 0. Repeated external-entity symbols represent the same actors/services shown in the context diagram.

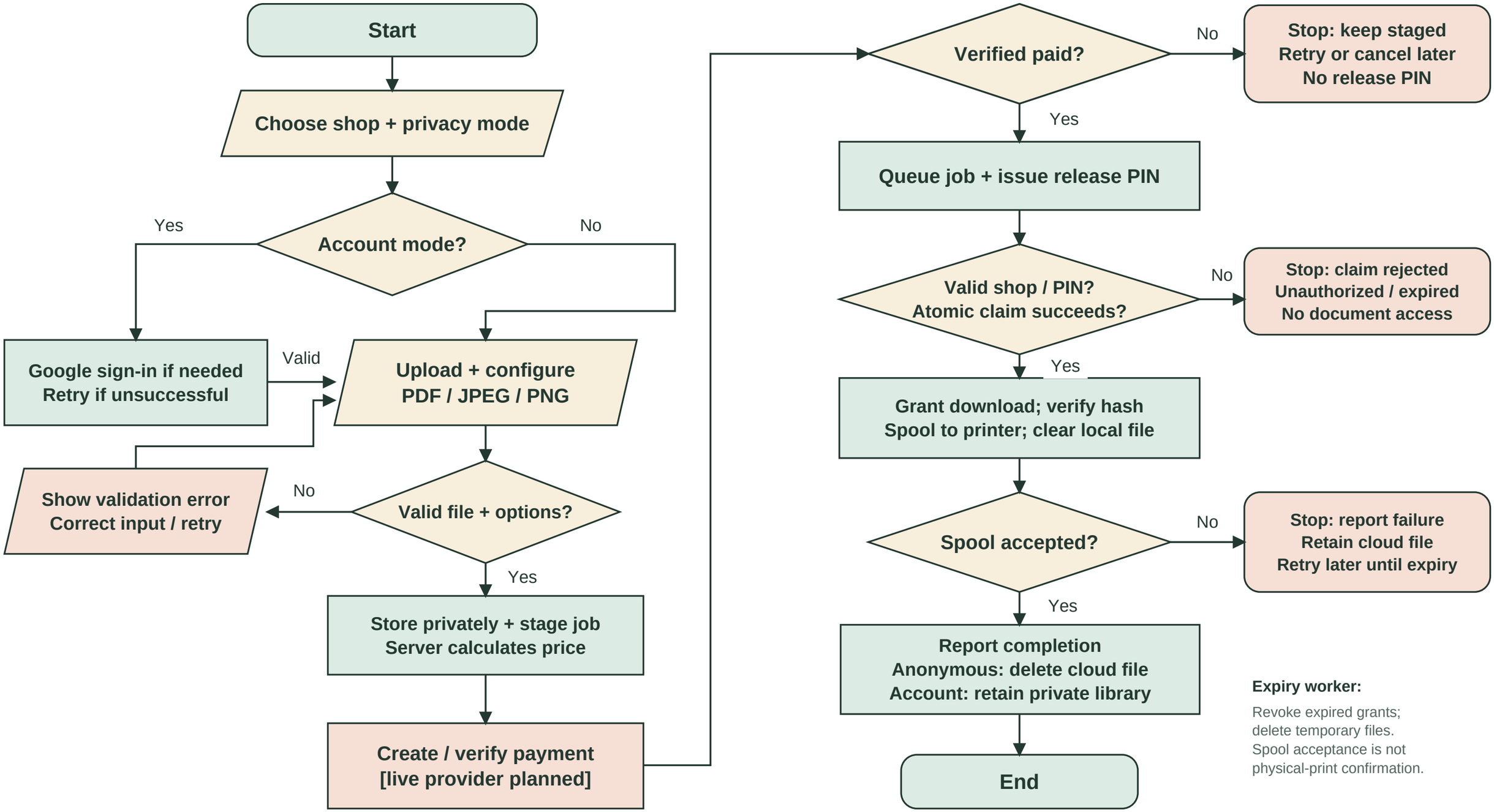


D2 is repeated for clarity: it is one logical group containing private files, job records, shop data and lifecycle audit information.

Staff sessions are independently authenticated. Store groups are not physical tables. Expiry scheduling is internal to process 5.0.

Control-flow diagram - submit, release and retain

Arrow direction is execution order. Every decision branch is labelled; failures do not silently become successful print jobs.

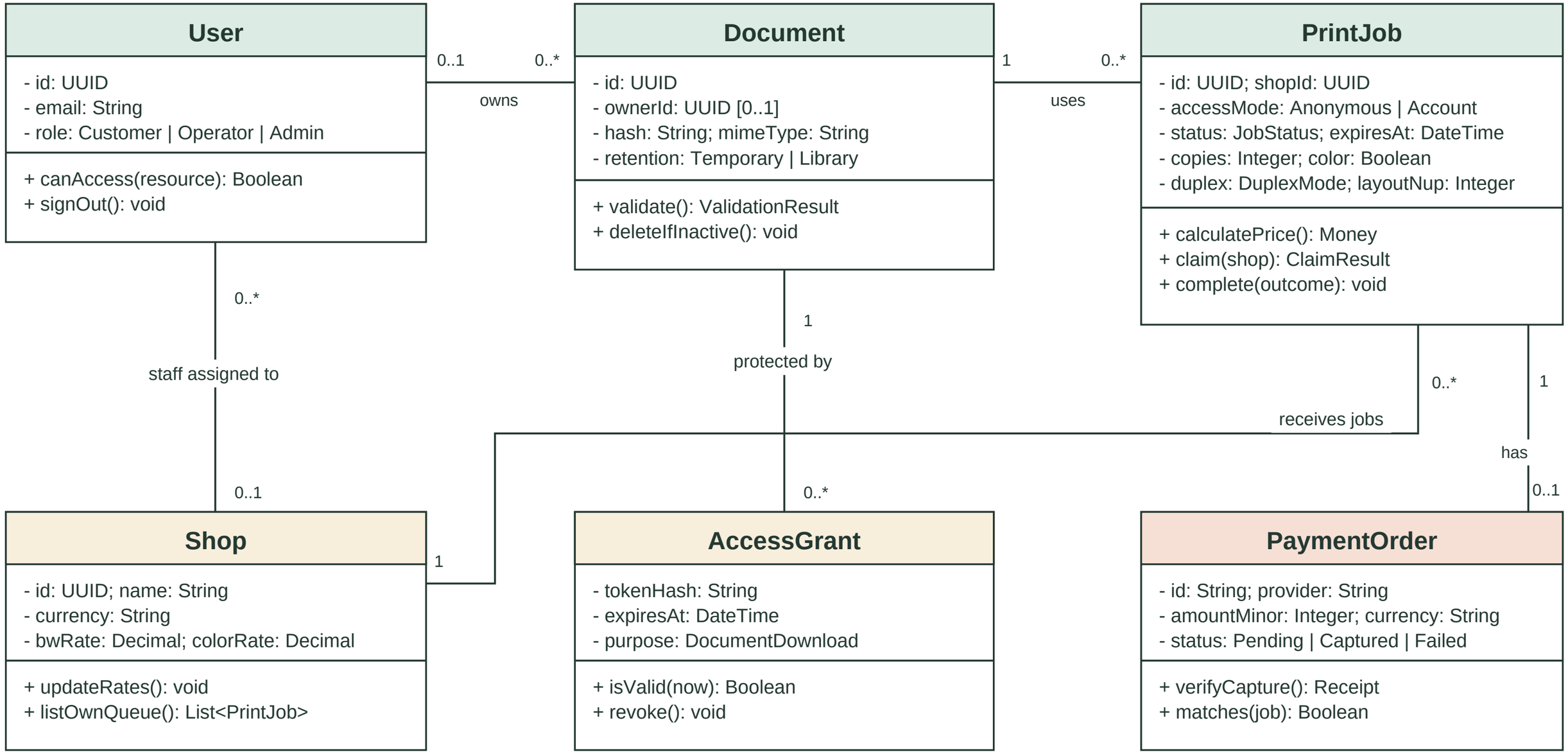


Expiry worker:

Revoke expired grants; delete temporary files. Spool acceptance is not physical-print confirmation.

UML class diagram - core domain

Conceptual object model: responsibilities and multiplicities, not the physical PostgreSQL table layout.

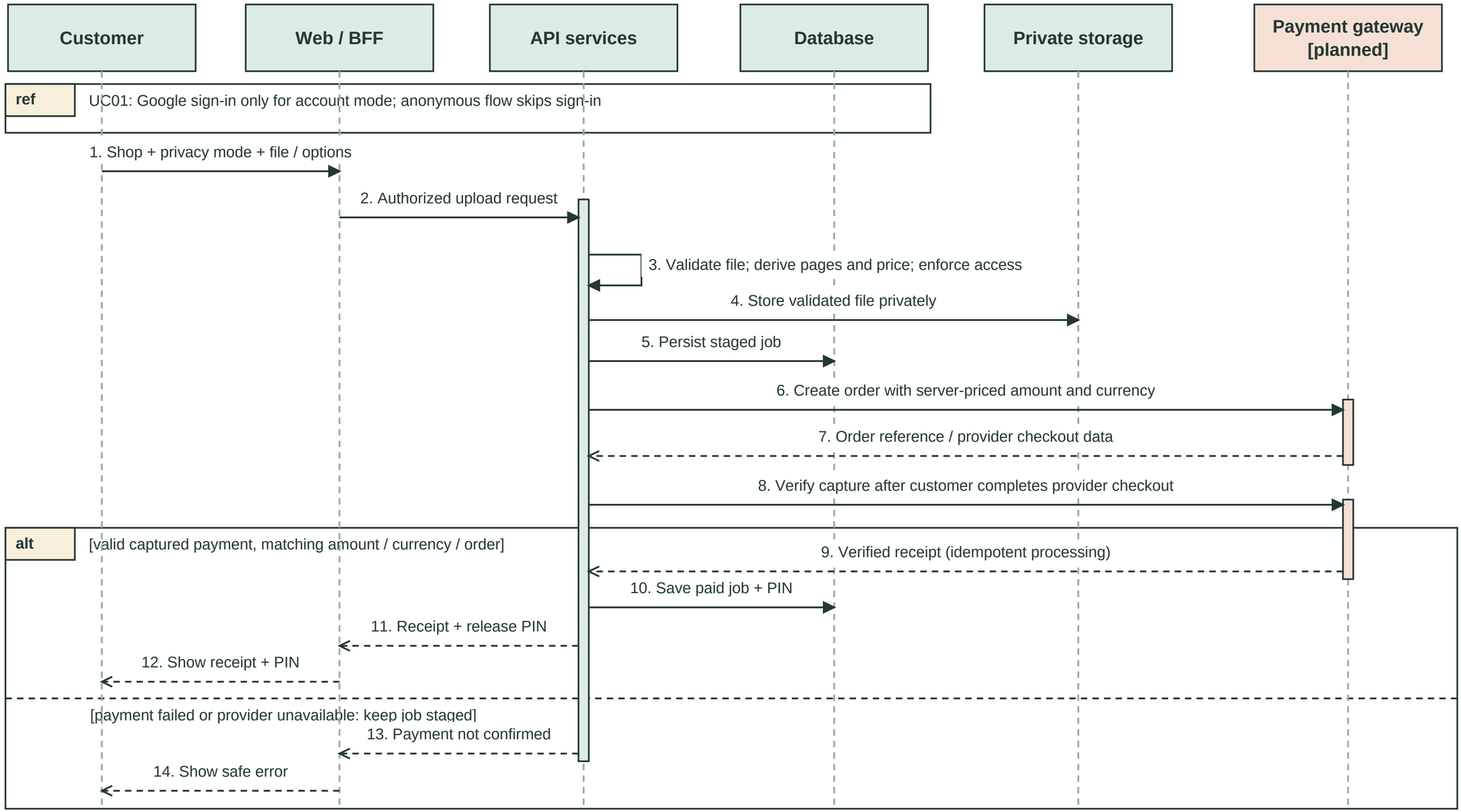


- private attribute + public operation | 0..1 optional 0..* many | Solid lines: associations

Constraints: anonymous document has no owner; each job uses one document at one shop. Live PaymentOrder adapter is planned.

UML sequence diagram - customer submission

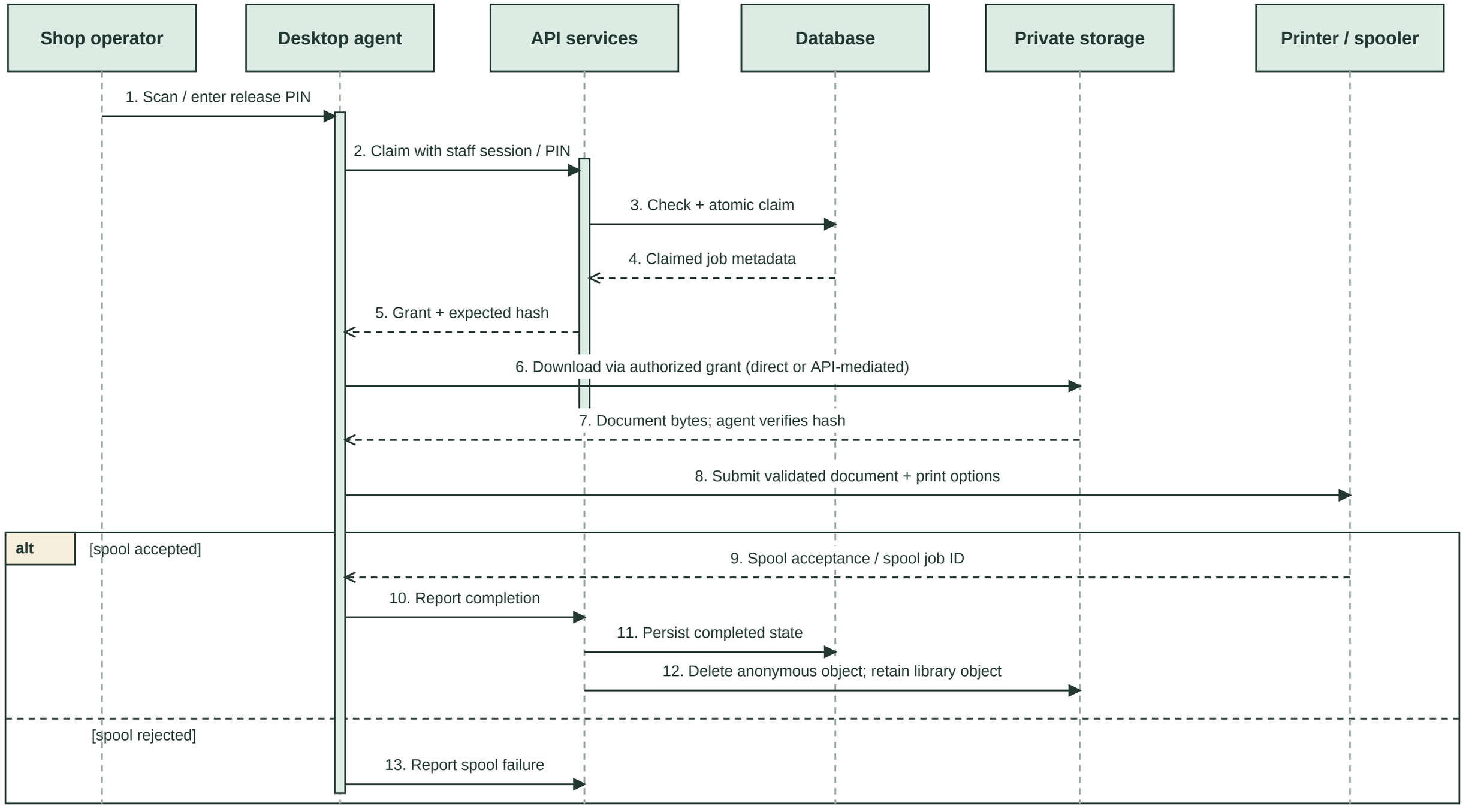
Scenario UC02. Authentication and server validation precede payment; the browser cannot mark a job as paid.



Solid arrow: request/call. Dashed open arrow: response. Narrow bar: activation. alt: alternative outcomes. ref: referenced interaction.

UML sequence diagram - print release and cleanup

Scenario UC07. A paid, unexpired job is released only within its assigned shop; documents are never public.



On failure: retain cloud file until expiry. Download/hash errors abort before step 8. Always clear local files. Spooling is not physical printing.